

Project 2: Predicting SAT Scores Using CRISP-DM Framework + Generative AI

CSCI 362 Database Management Systems

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Step 1.1 — Variable Identification via GenAI

Initial prompt:

"I am building a predictive model to estimate SAT scores for high school students. I have an initial dataset with variables such as number of AP test takes; number of tests taken; number of AP test passed. What additional independent variables might improve the model's predictive accuracy? For each suggested variable, explain the theoretical or empirical rationale for its inclusion. Lastly, group variables by category you deem most respectable"

Output:

1. Academic Performance Variables

These variables directly measure a student’s academic ability and preparedness for standardized testing.

Variable	Rationale for Inclusion
GPA (weighted/unweighted)	GPA is one of the strongest indicators of overall academic performance and long-term learning consistency. Students with higher GPAs often perform better on standardized exams due to stronger foundational knowledge and study habits.
PSAT Score	The PSAT closely mirrors the SAT in structure and content, making it a highly predictive variable for future SAT performance.
Honors Courses Taken	Enrollment in honors-level coursework may indicate academic rigor, motivation, and exposure to advanced material relevant to the SAT.
Attendance Rate	Students with higher attendance generally receive more instructional time and are less likely to fall behind academically.
Class Rank	Relative academic standing among peers may capture performance differences not fully reflected through GPA alone.
English and Math Course Grades	Since the SAT heavily evaluates math and reading/writing skills, subject-specific grades may improve prediction accuracy more than cumulative GPA alone.

2. Study Behavior and Test Preparation Variables

These variables capture preparation intensity and learning behaviors associated with standardized testing success.

Variable	Rationale for Inclusion
SAT Practice Hours	Students who dedicate more time to SAT preparation are more likely to improve familiarity with question types and testing strategies.
SAT Prep Course Participation	Formal test preparation programs may provide targeted instruction and strategy development that positively impacts scores.

Number of Practice Tests Taken	Repeated exposure to SAT-style exams can improve pacing, confidence, and test performance.
Homework Completion Rate	Consistent completion of assignments may reflect discipline and academic engagement.
Study Time Per Week	General study habits are often associated with stronger academic outcomes across subjects.

3. Socioeconomic and Environmental Variables

These variables represent external factors that may influence educational opportunities and academic performance.

Variable	Rationale for Inclusion
Household Income	Access to tutoring, prep courses, educational resources, and stable learning environments often correlates with income level.
Parent Education Level	Research frequently shows that students whose parents completed higher levels of education tend to perform better academically.
Internet Access at Home	Reliable internet access may affect access to learning materials, online practice exams, and virtual tutoring resources.
Access to Private Tutoring	Individualized instruction can significantly improve SAT preparation and subject mastery.
Hours Worked at a Job	Students working long hours outside school may have less time available for studying and SAT preparation.

4. Demographic and Student Background Variables

These variables may help identify broader patterns in educational outcomes.

Variable	Rationale for Inclusion
Grade Level	Junior and senior students may perform differently due to varying exposure to coursework and preparation.
English Language Learner (ELL) Status	Students still developing English proficiency may face additional challenges on reading and writing sections of the SAT.
Special Education Status	Certain accommodations or learning differences may influence standardized test outcomes.
Gender	Some studies have observed small performance differences across SAT sections, although this variable should be handled carefully to avoid bias.

5. School and Instructional Environment Variables

These variables evaluate how the educational environment itself may influence outcomes.

Variable	Rationale for Inclusion
Student-Teacher Ratio	Smaller class sizes may allow for more individualized instruction and support.
School Funding Per Student	Better-funded schools may provide more academic resources and test preparation opportunities.
Teacher Experience Level	More experienced teachers may contribute to stronger instructional quality and academic outcomes.
Availability of AP Courses	Schools offering more advanced coursework may better prepare students for college-level assessments like the SAT.

Analysis:

Do you agree with all suggested variables? Overall, many variables were removed primarily due to concerns involving ethical fairness, data availability, redundancy with existing predictors, or weak direct relationships to individual SAT performance. This helped maintain a more focused deployable predictive model. While some of these variables, like socioeconomic status and demographics may improve predictive accuracy, they could also introduce fairness concerns and make the model less ethically appropriate for educational decision-making. School funding and teacher experience were also not included because they could be difficult to collect data for and may not contribute enough predictive value relative to the added complexity.

I chose variables from just the categories of *1. academic performance* and *2. study behavior/test preparation* because they are directly related to SAT performance, measurable using realistic school or student data, and supported by educational research. As a result, the final variable groups prioritized factors most connected to student academic ability and SAT preparation behaviors.

Final Prompt: “what if we just keep 1. academic performance variables and 2. study behavior and test preparation variables. i think those are enough to improve the models predictive accuracy”

Final Output:

1. Academic Performance Variables

- GPA
- PSAT score
- Honors courses taken
- Attendance rate
- Class rank
- English and math grades

2. Study Behavior and Test Preparation Variables

- SAT practice hours
- SAT prep course participation
- Number of practice tests taken
- Homework completion rate

- Weekly study time

Step 1.2 — Data Source Identification

Initial prompt: “using the final categories, For each variable, identify a specific, publicly available data source. Include the source name, URL if available, the relevant dataset or table, and any limitations of the data (e.g., geographic scope, update frequency, access restrictions)”

Output:

1. Academic Performance Variables

Variable	Source Name	URL	Relevant Dataset/Table	Limitations
GPA (weighted/unweighted)	National Center for Education Statistics (NCES)	NCES High School Transcript Study	High School Transcript Study (HSTS)	National-level data only; individual student records may require restricted access.
PSAT Score	College Board	PSAT/NMSQT Program Data	Student-level PSAT data is not fully public and may require school or institutional access.	
Honors Courses Taken	NCES Education Longitudinal Study	Course-Taking and Academic Records Tables	Some datasets are older and may not reflect current curriculum trends.	
Attendance Rate	U.S. Department of Education Civil Rights Data Collection	Chronic Absenteeism Data	Primarily school-level rather than student-level data.	
Class Rank	NCES High School Longitudinal Study	Student Academic Performance Tables	Certain detailed variables may require restricted-use licenses.	

English and Math Course Grades	NCES Education Longitudinal Study	Transcript and Achievement Tables	Data updates are infrequent and may contain historical cohorts only.
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2. Study Behavior and Test Preparation Variables

Variable	Source Name	URL	Relevant Dataset/Table	Limitations
SAT Practice Hours	Kaggle Student Performance Datasets	Student Study Habits Datasets		Data quality varies because many Kaggle datasets are user-uploaded and not officially validated.
SAT Prep Course Participation	College Board Research Reports	SAT Preparation and Participation Studies		Some reports summarize trends rather than provide downloadable raw datasets.
Number of Practice Tests Taken	Khan Academy Official SAT Practice	SAT Practice Usage Statistics		Detailed student-level data may not be publicly downloadable.
Homework Completion Rate	NCES Educational Longitudinal Studies	Student Engagement and Homework Tables		Self-reported data may introduce response bias.
Weekly Study Time	Programme for International Student Assessment (PISA)	Student Learning Time Questionnaires		International dataset; may not fully reflect U.S.-specific SAT preparation behaviors.

Overall Data Source Considerations

One issue I noticed while researching datasets was that many educational datasets were either too broad, outdated, or required institutional access to obtain student-level records. Because of this, I decided to focus on variables that consistently appeared across freely downloadable datasets and could realistically be collected by a school. I also noticed that many study-behavior variables relied on self-reported information, which may introduce some inaccuracy into the model. Even with these limitations, the selected datasets still provided enough academic and behavioral information to support meaningful SAT score prediction.

Second prompt: “is it possible to find freely downloadable datasets for my final categories”

Output:

Yes — and honestly that’s probably the best direction for your project.

Instead of relying on restricted institutional datasets, you can build the project around **freely downloadable educational datasets** that already contain variables closely matching your final categories:

- academic performance
- study behavior/test preparation

That makes your project:

- reproducible
- realistic
- easier to defend academically
- easier to implement later during modeling

Here are the strongest options you can realistically use.

Recommended Freely Downloadable Datasets

1. [Students Performance Dataset \(Kaggle\)](#)

This is probably your best overall option.

Variables Included

- GPA-like academic performance metrics
- study time
- homework/study habits
- attendance
- parental involvement
- extracurricular participation

Why It Works Well

- Freely downloadable
- Clean CSV format
- Already structured for machine learning
- Contains many variables aligned with your final categories
- Easy to use in Python/R

Limitations

- Not SAT-specific
- Some variables are self-reported
- Smaller sample size than national datasets

2. [OpenIntro GPA & Study Hours Dataset](#)

Good supplemental dataset for validating the relationship between GPA and study behavior.

Variables Included

- GPA
- weekly study hours

Why It Works Well

- Completely public
- Very easy to analyze
- Good for exploratory analysis and correlation testing

Limitations

- Small dataset
 - College students instead of high school students
 - Limited number of variables
-

3. [College Board SAT Practice Resources](#)

Useful as supporting evidence for test preparation variables.

Variables/Usefulness

- practice test participation
- SAT preparation context
- standardized testing structure

Why It Helps

Even though it is not a downloadable student dataset, it supports the theoretical reasoning behind including:

- practice tests
- prep behavior
- SAT study variables

Limitations

- No downloadable student-level data
- Primarily informational resources

4. [NCES Educational Longitudinal Studies](#)

Best source for educational credibility.

Variables Included

- GPA
- course grades
- attendance
- academic engagement
- homework/study habits

Why It Helps

- Government-backed educational datasets
- Widely cited in academic research
- Strong credibility for your report

Limitations

- Large and harder to clean
- Some detailed student variables may require restricted access
- Older cohorts

Due to institutional access limitations for official SAT student records, the project will rely on publicly available educational datasets containing academically relevant proxy variables such as GPA, attendance, study hours, and course performance. These datasets still allow meaningful predictive modeling while maintaining reproducibility and ethical accessibility. However since sat practice scores are unattainable, along with other variables, we will remove them.

Final Variables:

Academic Performance Variables

1. GPA
2. Attendance Rate
3. English/Math Course Grades

Study Behavior Variables

4. Homework Completion Rate
5. Weekly Study Time

Analysis:

Variable	Source Name	URL	Relevant Dataset/Table
GPA	UCI Student Performance Dataset	Student Performance Dataset (G3 final grade used as GPA proxy)	Portuguese secondary school dataset; grades are not U.S. GPA scales.
Attendance Rate	UCI Student Performance Dataset	absences variable	Measures absences rather than direct attendance percentage.
English/Math Course Grades	UCI Student Performance Dataset	G1, G2, G3 grade variables for Math and Portuguese courses	Dataset contains Portuguese language course performance instead of SAT English-specific scores.
Homework Completion Rate	Kaggle Student Habits vs Academic Performance Dataset	Homework and study habit variables	User-uploaded dataset; data quality may vary.
Weekly Study Time	UCI Student Performance Dataset	studytime variable	Study time grouped into ranges instead of exact weekly hours.

Step 3.1 — Data Cleaning via GenAI
Verification and Quality Assurance:

- No rows contained missing values
 - No inconsistent data types were detected
 - No outliers exceeded the $3 \times$ column mean threshold
- Therefore, no rows or cells required modification or removal

Model 1 — Full-Variable Model

Linear Regression Model Results

1. Regression Equation

$$\text{SAT_Score} = 0.0000 + 180.0000(\text{GPA}) + 2.0000(\text{Attendance_Rate}) + 3.0000(\text{English_Math_Grades}) + 1.5000(\text{Homework_Completion_Rate}) + 8.0000(\text{Weekly_Study_Time_Hours})$$

2. Model Performance Metrics

R²: 1.0000
Mean Squared Error (MSE): 0.0000

3. Variable Significance Testing

Variable	Coefficient	P-Value	Significant at $\alpha = 0.05$
GPA	180.0000	0.000000	Yes
Attendance_Rate	2.0000	0.000000	Yes
English_Math_Grades	3.0000	0.000000	Yes
Homework_Completion_Rate	1.5000	0.000000	Yes
Weekly_Study_Time_Hours	8.0000	0.000000	Yes

Model 2 — AI-Selected Feature Model

The feature-selected linear regression model has been completed using **LASSO regression** for feature selection.

Results Summary

- Feature Selection Method: **LASSO (Least Absolute Shrinkage and Selection Operator)**
- Selected Variables:
 - GPA
 - Attendance_Rate
 - English_Math_Grades
 - Homework_Completion_Rate
 - Weekly_Study_Time_Hours
- R² on training data: **1.0000**
- MSE on training data: **0.0000**

Feature Selection Linear Regression Report

1. Feature Selection Methodology

LASSO (Least Absolute Shrinkage and Selection Operator) regression was used for feature selection. LASSO applies a penalty to regression coefficients, shrinking less important variables toward zero. Variables with coefficients reduced to zero were excluded from the final regression model. This method was selected because it helps reduce overfitting, improves interpretability, and automatically identifies the strongest predictors.

2. Selected Variables

Variable	Reason for Selection
GPA	Retained by LASSO due to predictive contribution to SAT_Score.
Attendance_Rate	Retained by LASSO due to predictive contribution to SAT_Score.
English_Math_Grades	Retained by LASSO due to predictive contribution to SAT_Score.
Homework_Completion_Rate	Retained by LASSO due to predictive contribution to SAT_Score.
Weekly_Study_Time_Hours	Retained by LASSO due to predictive contribution to SAT_Score.

3. Regression Equation

$$\text{SAT_Score} = 0.0000 + 180.0000(\text{GPA}) + 2.0000(\text{Attendance_Rate}) + 3.0000(\text{English_Math_Grades}) + 1.5000(\text{Homework_Completion_Rate}) + 8.0000(\text{Weekly_Study_Time_Hours})$$

4. Model Performance

R² on training data: 1.0000

MSE on training data: 0.0000

Model Comparison Analysis (Required Written Response)

6. Variables Used in Each Model

Model 1 included every available predictor variable:

- GPA
- Attendance_Rate
- English_Math_Grades
- Homework_Completion_Rate
- Weekly_Study_Time_Hours

Model 2 used LASSO feature selection to determine which variables contributed meaningful predictive value to the model. After applying LASSO, the same five variables were retained:

- GPA
- Attendance_Rate
- English_Math_Grades
- Homework_Completion_Rate
- Weekly_Study_Time_Hours

Model 1 used all available independent variables in the dataset, while Model 2 applied LASSO (Least Absolute Shrinkage and Selection Operator) regression as a feature selection method before fitting the final regression model.

7.

Model Performance Comparison			
Model	Variables Used	R ²	MSE (Training)
Model 1 – Full Regression	GPA, Attendance_Rate, English_Math_Grades, Homework_Completion_Rate, Weekly_Study_Time_Hours	1.0000	0.0000
Model 2 – LASSO Feature Selection Regression	GPA, Attendance_Rate, English_Math_Grades, Homework_Completion_Rate, Weekly_Study_Time_Hours	1.0000	0.0000

8. Model 2 would be my preferred model. Even though both models ended up with the same R² and MSE values, I think Model 2 is stronger because it actually tests whether each variable deserves to be included instead of automatically using every available predictor. By using LASSO feature selection, the model evaluates which variables contribute meaningful predictive value and helps reduce unnecessary complexity.

Evaluation Questions

9.

Model 1 Test MSE: 0.0000

Model 2 Test MSE: 0.0000

10.

Model	Variables Used	R ² (Train)	MSE (Train)	MSE (Test)
Model 1 – Full Regression	GPA, Attendance_Rate, English_Math_Grades, Homework_Completion_Rate, Weekly_Study_Time_Hours	1.0000	0.0000	0.0000
Model 2 – LASSO Feature Selection Regression	GPA, Attendance_Rate, English_Math_Grades, Homework_Completion_Rate, Weekly_Study_Time_Hours	1.0000	0.0000	0.0000

11.

My model preference did not change after testing. I still prefer Model 2 because it uses LASSO feature selection to confirm that each variable actually contributes to predicting SAT scores instead of automatically including every variable. Even though both models had identical R² and MSE values, Model 2 feels more methodologically sound and realistic from a data science perspective. I also did not observe major signs of overfitting since the training and testing MSE values remained the same across both datasets.

12.

The final regression equation is:

$$\text{SAT_Score} = 0.0000 + 180.0000(\text{GPA}) + 2.0000(\text{Attendance_Rate}) + 3.0000(\text{English_Math_Grades}) + 1.5000(\text{Homework_Completion_Rate}) + 8.0000(\text{Weekly_Study_Time_Hours})$$

This equation indicates that GPA had the strongest influence on predicted SAT scores, followed by English/Math grades and weekly study time. Attendance rate and homework completion rate also contributed positively to the prediction.

Several limitations should still be acknowledged. The model was trained on a relatively small dataset and included proxy variables rather than official SAT preparation or institutional student records. Despite these limitations, the project successfully demonstrated the full CRISP-DM process for predictive modeling and evaluation.

Step 6: Deployment

Deployment Deliverables

Executive Summary	The SAT score prediction model was developed to help Zootopia High School identify patterns between student academic behaviors and likely SAT performance. The model uses variables such as GPA, attendance rate, English and math grades, homework completion rate, and weekly study time to estimate a student's SAT score. The final model demonstrated very strong predictive performance during both training and testing, achieving an R^2 value of 1.0000 and an MSE of 0.0000 on the available dataset. While these results indicate extremely high accuracy within this project, the predictions should still be interpreted as estimates rather than guarantees. SAT performance can also be influenced by factors not included in the dataset, such as stress, personal circumstances, or test-day conditions. The school should use the model as a decision-support tool rather than a replacement for counselor judgment. Predictions can help identify students who may benefit from additional academic support or SAT preparation resources, but they should not be used to label students or make high-stakes academic decisions on their own.
Use Case Scenarios	One possible use case involves identifying students who may be academically capable but are at risk of underperforming on the SAT due to poor attendance or inconsistent study habits. For example, a counselor could use the model to flag students with strong grades but low predicted SAT scores and recommend tutoring, study workshops, or time-management support before testing season. Another use case involves resource allocation. School administrators could analyze groups of students with lower predicted SAT scores to determine where additional academic support programs should be focused. If many students with lower predictions also show low homework completion rates or limited study time, the school could expand after-school study programs or SAT preparation resources targeting those areas. The model could also assist counselors during college readiness planning by helping identify students who may need earlier intervention before college application deadlines approach.
Ethical Considerations	One concern with deploying this model is that the dataset may not fully represent every type of student. If the training data is limited or biased, the predictions may be less accurate for certain groups of students. To help reduce this issue, the school should regularly retrain the model using updated student data and review whether prediction accuracy changes across different student populations. Another concern is that students could feel discouraged if predicted SAT scores are treated too seriously. The model should be used only as a support tool for counselors rather than as a label for student ability. Predictions should be used to guide intervention and support, not to define a student's potential or future opportunities.

Maintenance Plan	The model should probably be retrained once every school year so it stays aligned with current student performance patterns and grading standards. A reevaluation would also be necessary if the SAT exam changes significantly, if the school changes curriculum requirements, or if there are noticeable demographic changes within the student population. Oversight should involve both technical staff and school counselors so the model stays accurate while still being used responsibly in real academic settings. Counselors should also regularly review whether the predictions are actually helping students and whether the model continues to provide useful results over time.
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Prompt–Response Submission

Step	Instructor Seed Prompt	My Final Prompt	GenAI Output Summary	My Analysis & Comments
1.1 Variable ID	Provide additional independent variables for SAT score prediction with rationale.	“what if we just keep 1. academic performance variables and 2. study behavior and test preparation variables. i think those are enough to improve the models predictive accuracy”	Keeping only: Academic Performance Variables, Study Behavior and Test Preparation Variables makes your model more focused, practical, and less vulnerable to ethical concerns or noisy/unavailable data.	The final variable groups prioritized factors most connected to student academic ability and SAT preparation behaviors.
1.2 Data Sources	Identify data sources for each selected variable.	“is it possible to find freely downloadable datasets for my final categories”		After evaluating dataset accessibility and feasibility, the final variable selection was narrowed to variables consistently available within publicly downloadable educational datasets. Several initially proposed variables, including PSAT scores and SAT preparation participation, were removed due to limited public accessibility. This improved the scalability

				of the proposed predictive model.
2–3 Data Cleaning	Clean the dataset per specified rules and show before/after.	with the final variable list Academic Performance, Variables, GPA, Attendance Rate, English/Math Course Grades, Study Behavior Variables, Homework, Completion Rate, Weekly Study Time identify a specific, publicly available data source. Include the source name, URL if available, the relevant dataset or table, and any limitations of the data data must be freely downloadable THEN Assemble all chosen variables into a single CSV or Excel file		the datasets originate from different populations and educational systems
3 Train/Test Split	Split data 60/40 with random seed.	Using the attached cleaned dataset, randomly partition the data into two files: • training_data.csv: 60% of records, selected at random. • testing_data.csv: the remaining 40% of records. Set a random seed of [insert a number, e.g., 42] to ensure reproducibility. Confirm the exact row counts and the random seed used in your response. Always specify a random seed so your split is reproducible. Name files exactly as shown: training_data.csv and testing_data.csv	The cleaned dataset was randomly partitioned using a reproducible random seed. Split Details • Random seed used: 42 • Total records: 100 • Training dataset (training_data.csv): 60 rows • Testing dataset (testing_data.csv): 40 rows	
4 Model 1	Fit linear regression using all variables; report equation, R^2 , MSE, pvalues.	Using the attached training_data.csv, fit a linear regression model to predict SAT_Score. Use all other columns in the dataset as independent variables. Report	The linear regression model was fitted using all available independent variables in training_data.csv	

		the following: 1. The regression equation (list each variable, its coefficient, and the intercept). 2. The R^2 (coefficient of determination). 3. The Mean Squared Error (MSE) on the training data. 4. For each independent variable, report its p-value and indicate whether it is statistically significant at $\alpha = 0.05$.	to predict SAT_Score. Model Results R^2 (Coefficient of Determination): 1.0000 - Mean Squared Error (MSE): 0.0000	
4 Model 2	Fit linear regression with AI-chosen variables.	Using the attached training_data.csv, fit a linear regression model to predict SAT_Score. You may choose which independent variables to include. Apply an appropriate feature selection method (e.g., stepwise selection, LASSO, or correlation screening). Report: 1. Which variables were selected and why. 2. The regression equation. 3. R^2 on the training data. 4. MSE on the training data. explain the feature selection methodology	The feature-selected linear regression model has been completed using LASSO regression for feature selection. Results Summary - Feature Selection Method: LASSO (Least Absolute Shrinkage and Selection Operator) - Selected Variables: - GPA - Attendance_Rate - English_Math_Grades - Homework_Completion_Rate - Weekly_Study_Time_Hours R^2 on training data: 1.0000 - MSE on training data: 0.0000	GPA had the strongest coefficient in the final model, suggesting overall academic consistency was the most influential predictor of SAT performance within the dataset. This makes sense because GPA reflects long-term academic consistency rather than short-term test preparation alone.
5 Evaluation	Compute MSE on test data for both models.		Model 1 Test MSE: 0.0000	My model preference did not change after testing. I

			Model 2 Test MSE: 0.0000	still prefer Model 2 because it uses LASSO feature selection to confirm that each variable actually contributes to predicting SAT scores instead of automatically including every variable. Even though both models had identical R ² and MSE values, Model 2 feels more methodologically sound and realistic from a data science perspective. I also did not observe major signs of overfitting since the training and testing MSE values remained the same across both datasets. The unusually perfect performance metrics likely reflect the simplified and partially synthetic nature of the dataset rather than realistic SAT prediction performance in a real educational environment.
6 Deployment	N/A (open-ended)			